

Spaces with symmetry and their (co)homology

Adam Pratt

University of Illinois at Chicago

April 4, 2022

G-Spaces

- In equivariant homotopy, study topological spaces with symmetries

G-Spaces

- In equivariant homotopy, study topological spaces with symmetries
- Mathematical notion of symmetry is neatly packaged into the structure of a group

G-Spaces

- In equivariant homotopy, study topological spaces with symmetries
- Mathematical notion of symmetry is neatly packaged into the structure of a group
- Consider topological spaces with G -action

G-Spaces

- In equivariant homotopy, study topological spaces with symmetries
- Mathematical notion of symmetry is neatly packaged into the structure of a group
- Consider topological spaces with G -action
- Can model these as functors from the group G represented as a category to the category of topological spaces

G-Spaces

- In equivariant homotopy, study topological spaces with symmetries
- Mathematical notion of symmetry is neatly packaged into the structure of a group
- Consider topological spaces with G -action
- Can model these as functors from the group G represented as a category to the category of topological spaces
- Define $G\text{Top} := \text{Fun}(BG, \text{Top})$

G-CW Complexes

- What might a G -cell look like?

G-CW Complexes

- What might a G -cell look like?
- Built out of regular the normal building blocks of CW complexes (that is, D^{n+1} and S^n) and the fundamental building blocks of G -actions

G -CW Complexes

- What might a G -cell look like?
- Built out of regular the normal building blocks of CW complexes (that is, D^{n+1} and S^n) and the fundamental building blocks of G -actions
- These are transitive G -sets, or *orbits* G/H for $H \leq G$

G -CW Complexes

- What might a G -cell look like?
- Built out of regular the normal building blocks of CW complexes (that is, D^{n+1} and S^n) and the fundamental building blocks of G -actions
- These are transitive G -sets, or *orbits* G/H for $H \leq G$
- Will be of the form $G/H \times D^{n+1}$ and $G/H \times S^n$, where the actions on D^{n+1} and S^n are trivial.

G -CW Complexes

Now, to define G -CW complexes, we simply copy the definition of a regular CW complex with our new cells.

G -CW Complexes

Now, to define G -CW complexes, we simply copy the definition of a regular CW complex with our new cells.

Definition

A G -CW complex is a sequential colimit of G -spaces X_n , where X_{n+1} is a pushout

$$\begin{array}{ccc} \coprod G/H \times S^n & \longrightarrow & X_n \\ \downarrow & & \downarrow \\ \coprod G/H \times D^{n+1} & \longrightarrow & X_{n+1}, \end{array}$$

where H varies over all closed subgroups of G .

Fixed points and homotopy groups

- Much like for any G -action, we must have fixed points of a G -space for any subgroup $H \leq G$

Fixed points and homotopy groups

- Much like for any G -action, we must have fixed points of a G -space for any subgroup $H \leq G$
- The *space of fixed points* X^H of a G -space X for a subgroup $H \leq G$ is given by $X^H := \text{map}(G/H, X)$

Fixed points and homotopy groups

- Much like for any G -action, we must have fixed points of a G -space for any subgroup $H \leq G$
- The *space of fixed points* X^H of a G -space X for a subgroup $H \leq G$ is given by $X^H := \text{map}(G/H, X)$
- Mapping out of cells gives a natural definition for homotopy groups, so define $\pi_n^H(X) := [G/H \times S^n, X]$

Fixed points and homotopy groups

- Much like for any G -action, we must have fixed points of a G -space for any subgroup $H \leq G$
- The *space of fixed points* X^H of a G -space X for a subgroup $H \leq G$ is given by $X^H := \text{map}(G/H, X)$
- Mapping out of cells gives a natural definition for homotopy groups, so define $\pi_n^H(X) := [G/H \times S^n, X]$
- By the tensor-hom adjunction, we have $\pi_n^H(X) = [S^n, \text{map}(G/H, X)] = \pi_n(X^H)$

Fixed points and homotopy groups

- Much like for any G -action, we must have fixed points of a G -space for any subgroup $H \leq G$
- The *space of fixed points* X^H of a G -space X for a subgroup $H \leq G$ is given by $X^H := \text{map}(G/H, X)$
- Mapping out of cells gives a natural definition for homotopy groups, so define $\pi_n^H(X) := [G/H \times S^n, X]$
- By the tensor-hom adjunction, we have $\pi_n^H(X) = [S^n, \text{map}(G/H, X)] = \pi_n(X^H)$
- So, homotopy groups are detected on fixed points

Coefficient systems

- Homotopy groups of a G -space are given by an abelian group for each subgroup of G

Coefficient systems

- Homotopy groups of a G -space are given by an abelian group for each subgroup of G
- However, they also remember the restriction maps $G/H \rightarrow G/K$ for $H \leq K$

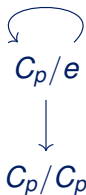
Coefficient systems

- Homotopy groups of a G -space are given by an abelian group for each subgroup of G
- However, they also remember the restriction maps $G/H \rightarrow G/K$ for $H \leq K$
- This data can be neatly packaged into a functor from a category known as the *orbit category* $\mathcal{O}_G$ of G , whose objects are orbits G/H and whose morphisms are such restriction maps

Coefficient systems

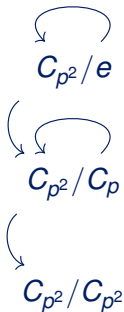
- Homotopy groups of a G -space are given by an abelian group for each subgroup of G
- However, they also remember the restriction maps $G/H \rightarrow G/K$ for $H \leq K$
- This data can be neatly packaged into a functor from a category known as the *orbit category* $\mathcal{O}_G$ of G , whose objects are orbits G/H and whose morphisms are such restriction maps
- These presheaves $\mathcal{O}_G^{\text{op}} \rightarrow \text{Ab}$ that classify the homotopy groups of a G -space are known as *coefficient systems*

The orbit category of C_p



A visual representation of the orbit category of the cyclic group C_p

The orbit category of C_{p^2}



A visual representation of the orbit category of the cyclic group C_{p^2}

Elmendorf's Theorem

Since homotopy groups of G -spaces are presheaves on the orbit category, our model of G -spaces as $\mathrm{Fun}(BG, \mathrm{Top})$ may not be the best model for the homotopy theory of G -spaces.

Elmendorf's Theorem

Since homotopy groups of G -spaces are presheaves on the orbit category, our model of G -spaces as $\mathrm{Fun}(BG, \mathrm{Top})$ may not be the best model for the homotopy theory of G -spaces.

A theorem of Elmendorf gives us a more suitable model:

Elmendorf's Theorem

The category of presheaves of spaces on the orbit category, $\mathrm{Fun}(\mathcal{O}_G^{\mathrm{op}}, \mathrm{Top})$ has a homotopy theory equivalent to that of $G\mathrm{Top}$ where the weak equivalences are taken pointwise; that is, equivalences are detected by the homotopy groups we defined above.

Example: Beach Ball



- Consider a beach ball with six vertical sections and let C_6 act by rotation about the y -axis

Example: Beach Ball



- Consider a beach ball with six vertical sections and let C_6 act by rotation about the y -axis
- Since the north and south poles are fixed by all of C_6 , they give two 0-cells $C_6/C_6 \times D^0$

Example: Beach Ball



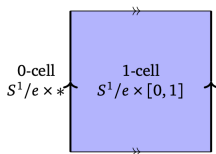
- Consider a beach ball with six vertical sections and let C_6 act by rotation about the y -axis
- Since the north and south poles are fixed by all of C_6 , they give two 0-cells $C_6/C_6 \times D^0$
- The six lines between the wedges are only fixed by the identity, so they give a single 1-cell $C_6/e \times D^1$

Example: Beach Ball



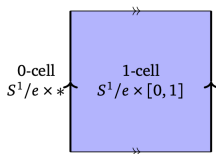
- Consider a beach ball with six vertical sections and let C_6 act by rotation about the y -axis
- Since the north and south poles are fixed by all of C_6 , they give two 0-cells $C_6/C_6 \times D^0$
- The six lines between the wedges are only fixed by the identity, so they give a single 1-cell $C_6/e \times D^1$
- The six wedges themselves are only fixed by the identity, so they give a single 2-cell $C_6/e \times D^2$

Example: Torus



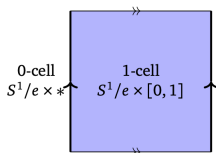
- Consider the torus $T = S^1 \times S^1$. Let S^1 act on T by multiplication on the first coordinate, that is, $z \cdot (z_1, z_2) = (zz_1, z_2)$

Example: Torus



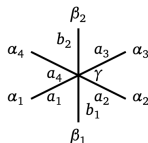
- Consider the torus $T = S^1 \times S^1$. Let S^1 act on T by multiplication on the first coordinate, that is, $z \cdot (z_1, z_2) = (zz_1, z_2)$
- No nontrivial fixed points, so we can equip it with the structure of a S^1 -CW complex with one 0-cell $S^1/e \times D^0$ and one 1-cell $S^1/e \times D^1$

Example: Torus



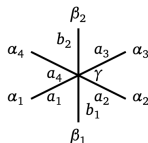
- Consider the torus $T = S^1 \times S^1$. Let S^1 act on T by multiplication on the first coordinate, that is, $z \cdot (z_1, z_2) = (zz_1, z_2)$
- No nontrivial fixed points, so we can equip it with the structure of a S^1 -CW complex with one 0-cell $S^1/e \times D^0$ and one 1-cell $S^1/e \times D^1$
- Only need 1-cells even though it is non-equivariantly 2-dimensional

Example: Star Graph



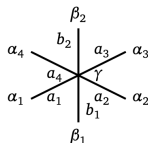
- Consider a star graph X above with one central vertex, six peripheral vertices, and six edges. Let C_4 act on X by sending $\alpha_i \mapsto \alpha_{i+1 \bmod 4}$ and $\beta_i \mapsto \beta_{i+1 \bmod 2}$, that is, C_4 rotates the α_i s and reflects the β_i s

Example: Star Graph



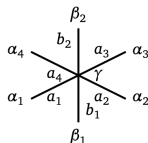
- Consider a star graph X above with one central vertex, six peripheral vertices, and six edges. Let C_4 act on X by sending $\alpha_i \mapsto \alpha_{i+1 \bmod 4}$ and $\beta_i \mapsto \beta_{i+1 \bmod 2}$, that is, C_4 rotates the α_i s and reflects the β_i s
- The center point γ is fixed by the whole group, so it gives a single 0-cell $C_4/C_4 \times D^0$

Example: Star Graph



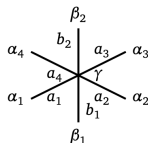
- Consider a star graph X above with one central vertex, six peripheral vertices, and six edges. Let C_4 act on X by sending $\alpha_i \mapsto \alpha_{i+1 \bmod 4}$ and $\beta_i \mapsto \beta_{i+1 \bmod 2}$, that is, C_4 rotates the α_i s and reflects the β_i s
- The center point γ is fixed by the whole group, so it gives a single 0-cell $C_4/C_4 \times D^0$
- The vertices β_i are fixed by the subgroup C_2 , so together they give a single 0-cell $C_4/C_2 \times D^0$

Example: Star Graph



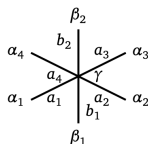
- The vertices α_j are only fixed by the identity, so together they give a single 0-cell $C_4/e \times D^0$

Example: Star Graph



- The vertices α_i are only fixed by the identity, so together they give a single 0-cell $C_4/e \times D^0$
- The edges b_i are also fixed by C_2 , so together they give a single 1-cell $C_4/C_2 \times D^1$

Example: Star Graph



- The vertices α_i are only fixed by the identity, so together they give a single 0-cell $C_4/e \times D^0$
- The edges b_i are also fixed by C_2 , so together they give a single 1-cell $C_4/C_2 \times D^1$
- The edges a_i are only fixed by the identity, so all together they give a single 1-cell $C_4/e \times D^1$

Equivariant Suspension

- When talking about cohomology, it is important to first discuss suspension

Equivariant Suspension

- When talking about cohomology, it is important to first discuss suspension
- Let V be a real representation of G , and let S^V be its one-point compactification

Equivariant Suspension

- When talking about cohomology, it is important to first discuss suspension
- Let V be a real representation of G , and let S^V be its one-point compactification
- V is a G -space in the obvious way, and S^V is a G -space where the point at infinity is a fixed point

Equivariant Suspension

- When talking about cohomology, it is important to first discuss suspension
- Let V be a real representation of G , and let S^V be its one-point compactification
- V is a G -space in the obvious way, and S^V is a G -space where the point at infinity is a fixed point
- Define the suspension $\Sigma^V X$ of a G -space X by $\Sigma^V X := S^V \wedge X$

Equivariant Suspension

- When talking about cohomology, it is important to first discuss suspension
- Let V be a real representation of G , and let S^V be its one-point compactification
- V is a G -space in the obvious way, and S^V is a G -space where the point at infinity is a fixed point
- Define the suspension $\Sigma^V X$ of a G -space X by $\Sigma^V X := S^V \wedge X$
- Really care about “stable maps”, that is, $\varinjlim_V \text{map}(\Sigma^V X, \Sigma^V Y)$

Wrong-way maps

- For $K \leq H \leq G$, we can embed H/K into some finite dimensional representation V

Wrong-way maps

- For $K \leq H \leq G$, we can embed H/K into some finite dimensional representation V
- let ν be the normal bundle of H/K , and let τ be the tangent bundle

Wrong-way maps

- For $K \leq H \leq G$, we can embed H/K into some finite dimensional representation V
- let ν be the normal bundle of H/K , and let τ be the tangent bundle
- We get a sequence of maps

$$S^V \rightarrow T(\nu) \rightarrow T(\nu \oplus \tau) \rightarrow S^V \wedge H/K_+,$$

Where T denotes the Thom construction.

Wrong-way maps

- For $K \leq H \leq G$, we can embed H/K into some finite dimensional representation V
- let ν be the normal bundle of H/K , and let τ be the tangent bundle
- We get a sequence of maps

$$S^V \rightarrow T(\nu) \rightarrow T(\nu \oplus \tau) \rightarrow S^V \wedge H/K_+,$$

Where T denotes the Thom construction.

- By induction (a functor we will define later), we get a map $G/H_+ \wedge S^V \rightarrow G/K_+ \wedge S^V$

Wrong-way maps

- For $K \leq H \leq G$, we can embed H/K into some finite dimensional representation V
- let ν be the normal bundle of H/K , and let τ be the tangent bundle
- We get a sequence of maps

$$S^V \rightarrow T(\nu) \rightarrow T(\nu \oplus \tau) \rightarrow S^V \wedge H/K_+,$$

Where T denotes the Thom construction.

- By induction (a functor we will define later), we get a map $G/H_+ \wedge S^V \rightarrow G/K_+ \wedge S^V$
- That is, stably, we get another map in the opposite direction of the restriction

Mackey Functors

- For cohomology, we need to think about a different category from the orbit category that keeps track of these “wrong-way maps”

Mackey Functors

- For cohomology, we need to think about a different category from the orbit category that keeps track of these “wrong-way maps”
- Adding in these transfer maps, we get a category known as the Burnside category, denoted $\mathcal{B}_G$

Mackey Functors

- For cohomology, we need to think about a different category from the orbit category that keeps track of these “wrong-way maps”
- Adding in these transfer maps, we get a category known as the Burnside category, denoted $\mathcal{B}_G$
- Additive functors from the Burnside category to $\mathbf{Ab}$ will be known as *Mackey functors*

Mackey Functors

- For cohomology, we need to think about a different category from the orbit category that keeps track of these “wrong-way maps”
- Adding in these transfer maps, we get a category known as the Burnside category, denoted $\mathcal{B}_G$
- Additive functors from the Burnside category to $\mathbf{Ab}$ will be known as *Mackey functors*
- Our cohomology will take values in the abelian category of Mackey functors instead of abelian groups

$RO(G)$ -grading

- Since we can suspend our spaces by any G -representation, it makes sense to grade our cohomology by representations instead of by integers

$RO(G)$ -grading

- Since we can suspend our spaces by any G -representation, it makes sense to grade our cohomology by representations instead of by integers

Definition

The *representation ring* of G , denoted $RO(G)$, is the ring whose underlying set is the set of isomorphism classes of finite-rank G -representations (over a particular ring R), with abelian group structure given by the Grothendieck group of the monoid of such representations under direct sum, and the multiplication given by the tensor product.

$RO(G)$ -grading

- Since we can suspend our spaces by any G -representation, it makes sense to grade our cohomology by representations instead of by integers

Definition

The *representation ring* of G , denoted $RO(G)$, is the ring whose underlying set is the set of isomorphism classes of finite-rank G -representations (over a particular ring R), with abelian group structure given by the Grothendieck group of the monoid of such representations under direct sum, and the multiplication given by the tensor product.

- Our cohomology theories will be graded by $RO(G)$

$\mathbb{Z}$ -graded theories

- We will always have $\mathbb{Z} \hookrightarrow RO(G)$, so one certainly does have $\mathbb{Z}$ -graded cohomology for G -spaces

$\mathbb{Z}$ -graded theories

- We will always have $\mathbb{Z} \hookrightarrow RO(G)$, so one certainly does have $\mathbb{Z}$ -graded cohomology for G -spaces
- They can be harder to compute at times since they have less structure

$\mathbb{Z}$ -graded theories

- We will always have $\mathbb{Z} \hookrightarrow RO(G)$, so one certainly does have $\mathbb{Z}$ -graded cohomology for G -spaces
- They can be harder to compute at times since they have less structure
- They lack some of the geometric information that $RO(G)$ -graded cohomology contains

$\mathbb{Z}$ -graded theories

- We will always have $\mathbb{Z} \hookrightarrow RO(G)$, so one certainly does have $\mathbb{Z}$ -graded cohomology for G -spaces
- They can be harder to compute at times since they have less structure
- They lack some of the geometric information that $RO(G)$ -graded cohomology contains
- There are some advantages to $\mathbb{Z}$ -graded theories as well though; in particular, $\mathbb{Z}$ -graded stable homotopy groups give the t -structure for equivariant stable homotopy theory

The cellular chain complex

- Let $\underline{[X, Y]}_G$ be the Mackey functor defined by

$$\underline{[X, Y]}_G(G/H) = [G/H_+ \wedge X, Y]_G.$$

The cellular chain complex

- Let $\underline{[X, Y]}_G$ be the Mackey functor defined by

$$\underline{[X, Y]}_G(G/H) = [G/H_+ \wedge X, Y]_G.$$

- Define a chain complex $\underline{C}_*^V(X)$ of Mackey functors as follows:

$$\underline{C}_n^V(X) = \underline{[S^{V-|V|+n}, X^n/X^{n-1}]}_G.$$

The cellular chain complex

- Let $\underline{[X, Y]}_G$ be the Mackey functor defined by

$$\underline{[X, Y]}_G(G/H) = [G/H_+ \wedge X, Y]_G.$$

- Define a chain complex $\underline{C}_*^V(X)$ of Mackey functors as follows:

$$\underline{C}_n^V(X) = \underline{[S^{V-|V|+n}, X^n/X^{n-1}]}_G.$$

- The differential $d_n : \underline{C}_n^V(X) \rightarrow \underline{C}_{n-1}^V(X)$ is given by the stable connecting homomorphism of the triple (X^n, X^{n-1}, X^{n-2})

Ordinary (co)homology

Now we can give the actual definition of $RO(G)$ -graded, Mackey functor-valued equivariant (co)homology:

Ordinary (co)homology

Now we can give the actual definition of $RO(G)$ -graded, Mackey functor-valued equivariant (co)homology:

Definition

The *ordinary cohomology* with coefficients in a Mackey functor M is given by

$$H_G^V(X; M) = H^{|V|}(\underline{\mathrm{Hom}}_{\mathcal{B}_G}(\underline{C}_*^V(X), M)).$$

The *ordinary homology* with coefficients in a Mackey functor M is given by

$$\underline{H}_V^G(X; M) = H_{|V|}(\underline{C}_*^V(X) \square M).$$

Where $\underline{\mathrm{Hom}}_{\mathcal{B}_G}$ denotes the internal hom in the category of Mackey functors and $\square$ denotes the symmetric monoidal product in the category of Mackey functors (given by the tensor product of functors, known as the Day convolution).

Induction and the Wirthmüller isomorphism

- Now we will compute a simple example of $RO(G)$ -graded cohomology with values in Mackey functors

Induction and the Wirthmüller isomorphism

- Now we will compute a simple example of $RO(G)$ -graded cohomology with values in Mackey functors
- These computations can get very complicated

Induction and the Wirthmüller isomorphism

- Now we will compute a simple example of $RO(G)$ -graded cohomology with values in Mackey functors
- These computations can get very complicated
- We will need one quick tool briefly mentioned earlier:

Induction and the Wirthmüller isomorphism

- Now we will compute a simple example of $RO(G)$ -graded cohomology with values in Mackey functors
- These computations can get very complicated
- We will need one quick tool briefly mentioned earlier:

Definition

The forgetful functor $U : G\text{Top} \rightarrow H\text{Top}$ for $H \leq G$ has a left adjoint $G \times_H - : H\text{Top} \rightarrow G\text{Top}$, where $G \times_H X = G \times X / \sim$, with $(gh, x) \sim (g, hx)$. This left adjoint is called the *induction* functor.

Induction and the Wirthmüller isomorphism

- Now we will compute a simple example of $RO(G)$ -graded cohomology with values in Mackey functors
- These computations can get very complicated
- We will need one quick tool briefly mentioned earlier:

Definition

The forgetful functor $U : G\text{Top} \rightarrow H\text{Top}$ for $H \leq G$ has a left adjoint $G \times_H - : H\text{Top} \rightarrow G\text{Top}$, where $G \times_H X = G \times X / \sim$, with $(gh, x) \sim (g, hx)$. This left adjoint is called the *induction* functor.

- Stably, induction is also a right adjoint for the forgetful functor. This statement is known as the *Wirthmüller isomorphism*

Example: C_{2+}

- We will compute the (co)homology of C_{2+} as a C_2 -space with coefficients in the constant Mackey functor $\{G/H \mapsto \mathbb{Z}\}$, denoted $\underline{\mathbb{Z}}$

Example: C_{2+}

- We will compute the (co)homology of C_{2+} as a C_2 -space with coefficients in the constant Mackey functor $\{G/H \mapsto \mathbb{Z}\}$, denoted $\underline{\mathbb{Z}}$
- This space has a very simple cell structure: a single 0-cell $C_2/e_+ \wedge D^0$

Example: C_{2+}

- We will compute the (co)homology of C_{2+} as a C_2 -space with coefficients in the constant Mackey functor $\{G/H \mapsto \mathbb{Z}\}$, denoted $\underline{\mathbb{Z}}$
- This space has a very simple cell structure: a single 0-cell $C_2/e_+ \wedge D^0$
- Note that $RO(C_2) \cong \mathbb{Z} \oplus \mathbb{Z}\sigma$, where σ is the sign representation, so we can simply bigrade our (co)homology

$$\underline{C}_*^V(C_{2+})$$

- We will begin by computing $\underline{C}_*^V(C_{2+}) = \underline{[S^{V-|V|+n}, C_{2+}^n / C_{2+}^{n-1}]}_{C_2}$

$$\underline{C}_*^V(C_{2+})$$

- We will begin by computing $\underline{C}_*^V(C_{2+}) = \underline{[S^{V-|V|+n}, C_{2+}^n / C_{2+}^{n-1}]}_{C_2}$
- Clearly this chain complex is concentrated in degree 0, so we are reduced to computing $\underline{[S^{p+\sigma q-(p+q)}, C_{2+}]}_{C_2}$

$$\underline{C}_*^V(C_{2+})$$

- We will begin by computing $\underline{C}_*^V(C_{2+}) = \underline{[S^{V-|V|+n}, C_{2+}^n / C_{2+}^{n-1}]}_{C_2}$
- Clearly this chain complex is concentrated in degree 0, so we are reduced to computing $\underline{[S^{p+\sigma q-(p+q)}, C_{2+}]}_{C_2}$
- We will compute this on orbits as follows:

- We will begin by computing $\underline{C}_*^V(C_{2+}) = \underline{[S^{V-|V|+n}, C_{2+}^n / C_{2+}^{n-1}]}_{C_2}$
- Clearly this chain complex is concentrated in degree 0, so we are reduced to computing $\underline{[S^{p+\sigma q-(p+q)}, C_{2+}]}_{C_2}$
- We will compute this on orbits as follows:
 - C_2/C_2 :

$$[S^{p+\sigma q-(p+q)}, C_{2+}]_{C_2} \cong [S^{p+q-(p+q)}, S^0] = [S^0, S^0] \cong \mathbb{Z},$$

where the first isomorphism comes from the fact that induction is right adjoint to the forgetful functor.

- C_2/e :

$$[S^{p+\sigma q-(p+q)} \wedge C_2/e_+, C_{2+}]_{C_2} \cong [S^0, C_{2+}] \cong [S^0, S^0 \vee S^0] \cong \mathbb{Z} \oplus \mathbb{Z},$$

where the first isomorphism comes from the fact that induction is left adjoint to the forgetful functor.

$$\underline{C}_*^V(C_{2+})$$

So, we have

$$\underline{C}_*^V(C_{2+}) \cong \begin{array}{c} \xrightarrow{\quad} \\ \mathbb{Z} \oplus \mathbb{Z} \\ \xleftarrow{\quad} \\ \mathbb{Z} \end{array}$$

in degree 0 and 0 in other degrees.

Ends and coends

- To compute our (co)homology, we will need certain weighted (co)limits called (co)ends

Ends and coends

- To compute our (co)homology, we will need certain weighted (co)limits called (co)ends
- A familiar example of a coend from algebraic topology is the geometric realization of a simplicial set

Ends and coends

- To compute our (co)homology, we will need certain weighted (co)limits called (co)ends
- A familiar example of a coend from algebraic topology is the geometric realization of a simplicial set
- Given a simplicial set $X_\bullet$, its *geometric realization* is a topological space $|X_\bullet|$ defined by:

$$|X_\bullet| = \int^{[n] \in \Delta} \Delta_{\text{Top}}^n \times X_n$$

$$\underline{H}_{C_2}^V(C_{2+}; \underline{\mathbb{Z}})$$

- The internal hom in functor categories, in particular Mackey functors, is computed using an end formula

$$\underline{H}_{C_2}^V(C_{2+}; \underline{\mathbb{Z}})$$

- The internal hom in functor categories, in particular Mackey functors, is computed using an end formula
- We have

$$\underline{\mathrm{Hom}}_{\mathcal{B}_{C_2}}(\underline{C}_*^V(C_{2+}), \underline{\mathbb{Z}}) \cong \int_{a \in \mathcal{B}_{C_2}} \mathrm{Hom}_{\mathrm{Ab}}(\underline{C}_*^V(C_{2+})(a), \mathbb{Z})$$

$H_{C_2}^V(C_{2+}; \mathbb{Z})$

- The internal hom in functor categories, in particular Mackey functors, is computed using an end formula
- We have

$$\underline{\mathrm{Hom}}_{\mathcal{B}_{C_2}}(\underline{C}_*^V(C_{2+}), \underline{\mathbb{Z}}) \cong \int_{a \in \mathcal{B}_{C_2}} \mathrm{Hom}_{\mathrm{Ab}}(\underline{C}_*^V(C_{2+})(a), \mathbb{Z})$$

- Since the end of two functors where one of the functors is constant is computed pointwise, we have

$$\underline{\mathrm{Hom}}_{\mathcal{B}_{C_2}}(\underline{C}_*^V(C_{2+}), \underline{\mathbb{Z}}) \cong \begin{array}{c} \curvearrowright \\ \mathbb{Z} \oplus \mathbb{Z} \\ \uparrow \quad \downarrow \\ \mathbb{Z} \end{array}$$

$$\underline{H}_{C_2}^V(C_{2+}; \underline{\mathbb{Z}})$$

$H^{|V|}$ is nonzero if and only if $|V|$ is zero, that is, if $p + q = 0$. So, our cohomology is given by

$$\underline{H}_{C_2}^V(C_{2+}; \underline{\mathbb{Z}}) \cong \begin{cases} \begin{array}{c} \text{↺} \\ \mathbb{Z} \oplus \mathbb{Z} \\ \text{↻} \\ \mathbb{Z} \end{array} & p + q = 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\underline{H}_V^{C_2}(C_{2+}; \underline{\mathbb{Z}})$$

- The tensor product of Mackey functors is computed using a coend formula

$$\underline{H}_V^{C_2}(C_{2+}; \underline{\mathbb{Z}})$$

- The tensor product of Mackey functors is computed using a coend formula
- For homology, we have

$$\underline{C}_*^V(C_{2+}) \square \underline{\mathbb{Z}} \cong \int^{a \in \mathcal{B}_{C_2}} \underline{C}_*^V(C_{2+})(a) \otimes \underline{\mathbb{Z}}$$

$$\underline{H}_V^{C_2}(C_{2+}; \underline{\mathbb{Z}})$$

- The tensor product of Mackey functors is computed using a coend formula
- For homology, we have

$$\underline{C}_*^V(C_{2+}) \square \underline{\mathbb{Z}} \cong \int^{a \in \mathcal{B}_{C_2}} \underline{C}_*^V(C_{2+})(a) \otimes \mathbb{Z}$$

- Again, the coend of two functors where one is constant is computed pointwise, so we have

$$\underline{C}_*^V(C_{2+}) \square \underline{\mathbb{Z}} \cong \begin{array}{c} \text{---} \curvearrowright \text{---} \\ \mathbb{Z} \oplus \mathbb{Z} \\ \uparrow \quad \downarrow \\ \mathbb{Z} \end{array}$$

$$\underline{H}_V^{C_2}(C_{2+}; \underline{\mathbb{Z}})$$

As before, our homology will vanish outside dimension 0, so we get

$$\underline{H}_V^{C_2}(C_{2+}; \underline{\mathbb{Z}}) \cong \begin{cases} \begin{array}{c} \text{ } \end{array} & p+q=0 \\ 0 & \text{otherwise} \end{cases}$$

Spanier-Whitehead Duality of Orbits

- The fact that our homology and cohomology are isomorphic is no coincidence

Spanier-Whitehead Duality of Orbits

- The fact that our homology and cohomology are isomorphic is no coincidence
- Consequence of a very useful theorem in equivariant homotopy theory that is a corollary of the Wirthmüller isomorphism:

Spanier-Whitehead Duality of Orbits

- The fact that our homology and cohomology are isomorphic is no coincidence
- Consequence of a very useful theorem in equivariant homotopy theory that is a corollary of the Wirthmüller isomorphism:

Theorem

Orbits G/H are their own equivariant Spanier-Whitehead dual. That is,

$$\varinjlim_V \operatorname{map}(\Sigma^V G/H_+, \Sigma^V S^0) \simeq \varinjlim_V \Sigma^V G/H_+.$$